

William D. Frazer

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Earth and Planets Laboratory
Carnegie Science
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Education:

Yale University, New Haven CT, USA

Ph.D., Earth and Planetary Sciences

December 2024

M.Phil., Earth and Planetary Sciences

December 2021

Binghamton University, Binghamton NY, USA

B.S. *cum laude*, Geological Sciences: Geophysics

Spring 2019

Appointments:

Carnegie Postdoctoral Fellow

2024-

Earth and Planets Laboratory, Carnegie Science

Graduate Student Researcher

2019-2024

Department of Earth and Planetary Sciences, Yale University

Graduate Research Intern

Fall 2022

Yale Carbon Containment Lab, Yale University

Undergraduate Researcher

2016-2019

Geophysics and Remote Sensing Laboratory, Binghamton University

IRIS Intern

Summer 2018

IGPP, Scripps Institution of Oceanography, University of California San Diego

Publications:

10. Monsalve, G., Wagner, L.S., Avellaneda-Jiménez, D.S., **Frazer, W.D.**, Alzate, M.A., “**Flat-slab subduction and overriding plate architecture beneath the northwestern Andes revealed by receiver functions.**” *Journal of Geophysical Research: Solid Earth*, 131, e2026JB034380, 2026, <https://doi.org/10.1029/2026JB034380>

9. **Frazer, W.D.**, & Park, J. “**Thick underplating and buoyancy of the Bermuda swell.**” *Geophysical Research Letters*, 52, e2025GL118279, 2025 <https://doi.org/10.1029/2025GL118279>

8. Frazer, W.D., Park, J., & Link, F., “Crustal anisotropy as evidence for hydration of the Martian upper mantle.” *Physics of the Earth and Planetary Interiors*, **367**, 2025, <https://doi.org/10.1016/j.pepi.2025.107434>
7. Frazer, W.D., & Park, J. “No globally detectable seismic interfaces within Earth’s mantle transition zone: Evidence for basalt enrichment.” *Earth and Planetary Science Letters*, **641C**, 118838, 2024, <https://doi.org/10.1016/j.epsl.2024.118838>
6. Frazer, W.D., & Park, J. “High-resolution mid-mantle imaging with multiple-taper SS-precursor estimates.” *Geophysical Journal International*, Volume 233, Issue 2, May 2023, Pages 1356–1371, <https://doi.org/10.1093/gji/ggac491>
5. Frazer, W.D., & Korenaga, J. “Dynamic topography and the nature of deep thick plumes.” *Earth and Planetary Science Letters*, **578**, 117286, 2022. <https://doi.org/10.1016/j.epsl.2021.117286>
4. Frazer, W.D., Doran, A.K., & Laske, G., “Benchmarking Automated Rayleigh-Wave Arrival Angle Measurements for USArray Seismograms.” *Seismological Research Letters*, **93** (2A): 763–776, 2021. <https://doi.org/10.1785/0220210189>
3. Frazer, W.D., & Park, J., “Seismic evidence of mid-mantle water transport beneath the Yellowstone region.” *Geophysical Research Letters*, **48**, e2021GL095838, 2021. <https://doi.org/10.1029/2021GL095838>
2. Nikulin, A., de Smet, T., Baur, J., Frazer, W.D., & Abramowitz, J. “Detection and identification of remnant PFM-1 ‘Butterfly Mines’ with a UAV-based thermal-imaging protocol.” *Remote Sensing*, **10**(11), 1642, 2018. <https://doi.org/10.3390/rs10111672>
1. Nikulin, A., de Smet, T., Frazer, W.D., Baur, J., Abramowitz, J., Finan, D., Denara, S., Aglietti, N., & Campos, G. “Drones and “butterflies”: A low-cost UAV system for rapid detection and identification of unconventional minefields.” *Journal of Conventional Weapons Destruction*, **22.3**, 2018.

Publications in review, revision, and prep.

Frazer, W.D., “Discontiguous underplating beneath the Hawaiian Islands.” submitted to *Geophysical Research Letters*

Frazer, W.D., “Limited Seismic Detectability of Mantle Transition Zone Mineral Phase Changes.” in prep.

Awards:

Carnegie Postdoctoral Fellowship
Earth and Planets Laboratory, Carnegie Science

2024

Prize Teaching Fellowship 2023
Graduate School of Arts and Science, Yale University
Student nominated for outstanding performance and promise as a teacher (EPS 212 Oman Trip)

Excellence in Teaching Prize 2023
Department of Earth and Planetary Sciences, Yale University
For outstanding contributions to the teaching process (EPS 212 Oman Trip)

Glenn G. Bartle Class of '52 Award 2019
Department of Geological Sciences, Binghamton University
Awarded to an outstanding senior for academic excellence and service to the department

NSF GRFP Honorable Mention 2019
National Science Foundation

Distinguish Independent Work 2019
Department of Geological Sciences, Binghamton University
Awarded for completion of advanced research as an undergraduate

Provost's Award for Excellence in Undergraduate Research 2019
Binghamton University
Awarded to an undergraduate for excellence in research and scholarship

Invited Talks:

Geology and Geophysics departmental seminar, Texas A&M University February 2025

Earth Sciences departmental seminar, Binghamton University April 2023

Funding:

National Science Foundation – GEO SPSE “**An Investigation of Plume-less Intraplate Volcanism: Earthquake Detection, 3D Seismic Imaging, and Mineral Physics Beneath Bermuda,**” \$510,012, Pending

Outreach and Media

Interview with [Royal Gazette](#) on BEST network June 2026

[@CarnegiePlanets](#) Instagram takeover for BEST network service run May 2026
Week-long content shared on Earth and Planets Laboratory Instagram

“Bermuda: A Geologic Mystery,” Public Lecture: [YouTube Recording](#) May 2026
Neighborhood Lecture Series, Earth and Planets Laboratory, Carnegie Science

Interview with [Live Science](#) on Bermuda GRL Paper, coverage by [ABC news](#) December 2025

Public tours of Earth and Planets Laboratory Field Geophysics Laboratory

Fall 2025

Field and Ship Experience:

Lead Investigator: Bermuda Earthquake and Structure Test (BEST), Bermuda (~3 weeks+) 2025-
Co-chief Scientist: R/V Marcus G. Langseth, Eastern Pacific (~1 month) Fall 2025
 MUSICA deployment recovery, Colombian Andes (~3 weeks) Spring 2025
 The Rutgers and Yale Rapid Seismic Array deployment, NJ, USA (~2 days) Spring 2024
 New England Seismic Network (NEST), MA, NH, VT, USA (~2 weeks total) 2019-2022
 R/V Marcus G. Langseth, Off coast AK, USA (~3 weeks) Summer 2019
 R/V Robert Gordon Sproul, Off coast San Diego, CA, USA (~3 Days) Summer 2018
 Nantahala National Forest, NC, USA (~10 days) Summer 2018
 Poverty Point, LA, USA (~1 week) Summer 2017

Field Trip Leadership:

Lead TA Yale EPS 212, Global Tectonics Field Trip – Sicily, Italy (~10 days) Spring 2024
 Co-Lead Yale EPS Dept. Field Trip – Mainland Greece, Cyclades, and Crete (~15 days) Summer 2023
 Lead TA Yale EPS 212, Global Tectonics Field Trip – Samail Ophiolite, Oman (~10 days) Spring 2023
 Lead TA Yale EPS 212, Global Tectonics Field Trip – Death Valley, CA, USA (~7 days) Spring 2022
 Co-Lead Yale EPS Dept. Field Trip – Western, USA (~3 weeks) Summer 2021

Teaching:

TA, EPS 312, Structural Geology Fall 2022
 TA, EPS 210, Geology of North America Spring 2022
 TA, EPS 110, Dynamic Earth Fall 2020
 TA, G&G 100, Natural Disasters Fall 2019

Selected First Author Conference Presentations:

Frazer, W.D., Park, J. “Thick underplating and buoyancy of the Bermuda swell.” Invited Talk at GAC-MAC 2026

Frazer, W.D., Wagner, L., “Geophysical Constraints on the Lithosphere Beneath Plume-less Ocean Bathymetric Swells.” Presentation at 2025 AGU Fall Meeting

Frazer, W.D., “Global Observations of Melt-induced Low Velocity Zones Surrounding the Mantle Transition Zone.” Presentation at 2025 SSA Annual Meeting

Frazer, W.D., Park, J. “Seismic Detectability of Mantle Transition Zone Mineral Phase Changes.” Presentation at 2024 AGU Fall Meeting

Frazer, W.D., Park, J. “The Martian Lithosphere Beneath the Northern Lowlands May Be Anisotropic.” Talk at 2024 Mars Interior and Geophysics After InSight

Frazer, W.D., Park, J. “No Globally Detectable Seismic Interfaces within Earth’s Mantle Transition Zone: Evidence for Basalt Enrichment.” Poster at 2023 AGU Fall Meeting

Frazer, W.D., Park, J. “Does the 520-km discontinuity exist? / A global search for water in Earth’s mantle transition zone.” Poster at 2023 Interior of the Earth Gordon Research Conference

Frazer, W.D., Park, J. “Global High-Resolution mid-Mantle Imaging with Multiple-Taper SS-Precursor Estimates.” Poster at 2023 EGU General Assembly

Frazer, W.D., Park, J. “Global High-Resolution mid-Mantle Imaging with Multiple-Taper SS-Precursor Estimates.” Talk at 2022 AGU Fall Meeting

Frazer, W.D., Link, F., Wolf, J. “Improving Shear-wave Splitting Estimates with Multiple-taper Spectral Analysis.” Poster at 2022 AGU Fall Meeting

Frazer, W.D., Park, J. “Multiple-Taper Correlation to Detect Precursors to SS Earthquake Waves.” Talk at 2022 Multitaper Spectral Analysis Workshop at Banff Research Station

Frazer, W.D., Park, J. “Searching for Mid-Mantle Water with Multitaper-Correlation SS Precursors.” Talk at 2022 EGU General Assembly

Frazer, W.D., Park, J. “Searching for Mid-Mantle Water with Multitaper-Correlation SS Precursors.” Poster at 2021 AGU Fall Meeting

Frazer, W.D., Park, J. Chen, M. “Receiver Functions of the Transition Zone Below Yellowstone: Searching for Water.” Poster at 2020 AGU Fall Meeting

Frazer, W.D., Laske, G., Doran, A.K. “Surface-wave arrival angles and wave-propagation effects at the USArray Transportable Array.” Poster at 2018 AGU Fall Meeting

Frazer, W.D., Bourke, J.R., de Smet, T., Nikulin, A. “Geophysical Survey of Poverty Point UNESCO World Heritage Site Mound A.” Poster at 2017 AGU Fall Meeting

Service:

Reviewer: Nature Communications, Geophysical Journal International, Geophysical Research Letters, Journal of Geophysical Research: Solid Earth, G-Cubed, Seismica, National Science Foundation

Postdoctoral Representative

Carnegie Institution Postdoctoral Association

2026-

Peer mentor

Yale EPS Graduate Student Peer Mentoring Program

2023-2024

Co-convener, EGU

Physics-based earthquake modeling and engineering sessions

2023

Dana Club Co-President

2021-2022

Head of Yale EPS graduate student organization

Co-representative for Geophysics

2021-2022

Yale EPS Colloquium Committee

Computational Skills

Python, MATLAB, high-performance computing (Slurm), AxiSEM3D, UNIX
Familiar with Fortran, R, Generic Mapping Tools